RT 08.07.2008

$\omega_0^2 = \frac{c}{m}$ $\quad 2\delta = \frac{d}{m}$ $\quad u = F^*$

A2:

(a) $-m\ddot{y} + F - d\dot{y} - cy = 0 \quad \Rightarrow \dot{x} = \begin{bmatrix} \dot{y} \\ \dot{y} \\ F \end{bmatrix}^{\bullet} = \begin{bmatrix} 0 & 1 & 0 \\ -\omega_0^2 & -2\delta & 1/s \\ 0 & 0 & -1/F \end{bmatrix} \cdot \underline{x} + \begin{bmatrix} 0 \\ 0 \\ 1/F \end{bmatrix} u$

$v = [1 \quad 0 \quad 0] \; \underline{x}$

(c) $G(s) = \dfrac{\overbrace{1/F\cdot m}^{b_0}}{\underbrace{1}_{a_3}\cdot s^3 + \underbrace{(2\delta + 1/F)}_{a_2}s^2 + \underbrace{(\omega_0^2 + 2\delta/F)}_{a_1}s + \underbrace{\omega_0^2/F}_{a_0}}$

(d) $r_1 = \omega_{gr}^3 - \frac{c}{mT}$; $r_2 = 3\omega_{gr}^2 - \frac{c}{m} - \frac{d}{mT}$; $r_3 = 3\omega_{gr} - \frac{d}{m} - \frac{1}{T}$

$q_1 = \omega_{gr}^3 T^2 m$

RT 04.02.2008

A2:

(c) $a_0 = \omega_0^2$; $a_1 = 2\delta$; $b_0 = \omega_0^2$ $\quad r_1 = 0$; $r_2 = \sqrt{2}\,\omega_0 - 2\delta = \sqrt{\frac{2}{LC}} - \frac{1}{RC}$

$q_1 = 1$

A4:

$G_0(s) = K_0 \dfrac{1+sT_n}{sT_n} \cdot \dfrac{1}{1+sT_1} \qquad K_0 = K_p \cdot K_s$

$G_w(s) = \dfrac{G_0}{1+G_0} \cdot \dfrac{1}{1+sT_F} \qquad T_F = T_n$

$T_n = \dfrac{\sqrt{2}\,\omega_g T_1 - 1}{\omega_g^2 T_1} \qquad K_p = \dfrac{\sqrt{2}\,\omega_g T_1 - 1}{K_s}$

RT1 13.02.2009

A2:

$$M1: \dot{i}_1 = -\frac{1}{L} u_e + \frac{1}{L} u_c \quad (1)$$

$$M2: \dot{i}_2 = -\frac{R}{L} i_2 + \frac{1}{L} u_c \quad (2)$$

$$K_1: \dot{u}_c = \frac{1}{C} i_1 - \frac{1}{C} i_2 \quad (3)$$

(3) nach $I_1(s)$ umstellen und in (1) einsetzen:

$$s^2 C\, U_c(s) + s I_2(s) = -\frac{1}{L} U_c(s) + \frac{1}{L} U_e(s) \quad (1^*)$$

(2) nach $I_2(s)$ umstellen und in (1^*) einsetzen.

$$(s^2 C + \frac{2}{L}) U_c(s) = \frac{1}{L} U_a(s) + \frac{1}{L} U_e(s) \quad (2^*)$$

$U_e(s)$ in (3) einsetzen:

$$U_e(s) = (s \frac{L}{R} + 1) U_a(s) \quad (3^*)$$

$$(3^*) \text{ in } (2^*) \Rightarrow \frac{U_a(s)}{U_e(s)} = \frac{1/L}{s^3 \frac{LC}{R} + s^2 C + s\frac{2}{R} + \frac{1}{L}}$$

A1:

(c) $k_1 > 0$ mind. Grenzstabilität

$k_2 > 0$

$$Re\{N(s)\} \overset{!}{>} 0$$

A3: (a) stationär genau

$$\lim_{s \to 0} \left(s \cdot G_w(s) \right) \overset{muss}{=} 1 \quad \text{damit st. genau}$$

oder

$$\lim_{s \to 0} \left(s \cdot E(s) \right) \overset{muss}{=} 0 \quad - \text{"} -$$

Berechnungsmöglichkeiten

(b) $K_P = \frac{1}{2T_1}$

(c) $G(s) = \frac{1}{1 + sT_1}$ Rechnung: $G_w = \frac{X(s)}{W(s)} = G(s) \cdot \frac{X(s)}{\tilde{W}(s)}$

A(s)

muss weg!

$b_0 + b_1 s + b_2 s^2 ...$

A4:

(a) $T_1 = T_t \pm \frac{\sqrt{3} \cdot T_t}{2}$

würde zu Instabilität führen

(b) $G_F(s) = \frac{1}{1 + s\frac{-T_t}{2}}$

negative Zeiten nicht realisierbar und wenn $\frac{b_0 \cdots b_n s^n}{a_0 \cdots a_m s^m}$ $n > m$,

da diff. Verhalten